

Mini-Design Challenge I: A Personalized Learning Experience with AI Chatbot

My chatbot is named Curia. I am starting to think about how to integrate some version of it into a platform/ web-app I am developing for math students and educators. On my platform, there will not be text input available, but users will be able to "chat" with predetermined inputs (such as asking for help, answering about their grade level or experience, etc). I wanted to build a version to start thinking about how I want to prompt such a chatbot, even if the full version will not be embedded into my product.

I used the prompts below, partially inspired by a recent report by Elevenlabs on best practices for structure prompts for agentic AI. I started with a short, broader prompt but was not getting the results I wanted. The chatbot was given far too much information without being asked. By expanding and providing a more detailed prompt, I was able to get results closer to my goals.

I provided guidance for Curia to first ask about the user's level of understanding and adjust their prompts and level of challenge according to the user's grade level and familiarity with concepts (if they are an adult). I also provided guidance for how to support students or teachers with scaffolds and differentiated learning by asking follow-up questions and suggested strategies.